**Project Design Phase-II**
**Data Flow Diagram & User Stories**

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| Date | 28 September 2026 |
| Team ID | SWTID-2026-2598 |
| Project Name | AI BlogNest API |
| Maximum Marks | 4 Marks |

**Data Flow Diagram (Level 0)**

External Entities: API Client (User/Reader) and the Google Gemini AI service.

Process: The Express.js server (Auth, Blog, and AI controllers).

Data Store: MongoDB — Users collection and Blogs collection.

- API Client → [HTTP Request: register/login/CRUD/AI] → Express Server
- Express Server → [Read/Write User & Blog documents] → MongoDB
- Express Server → [Prompt: topic or content] → Gemini AI → [Generated text] → Express Server
- Express Server → [HTTP Response: JSON] → API Client

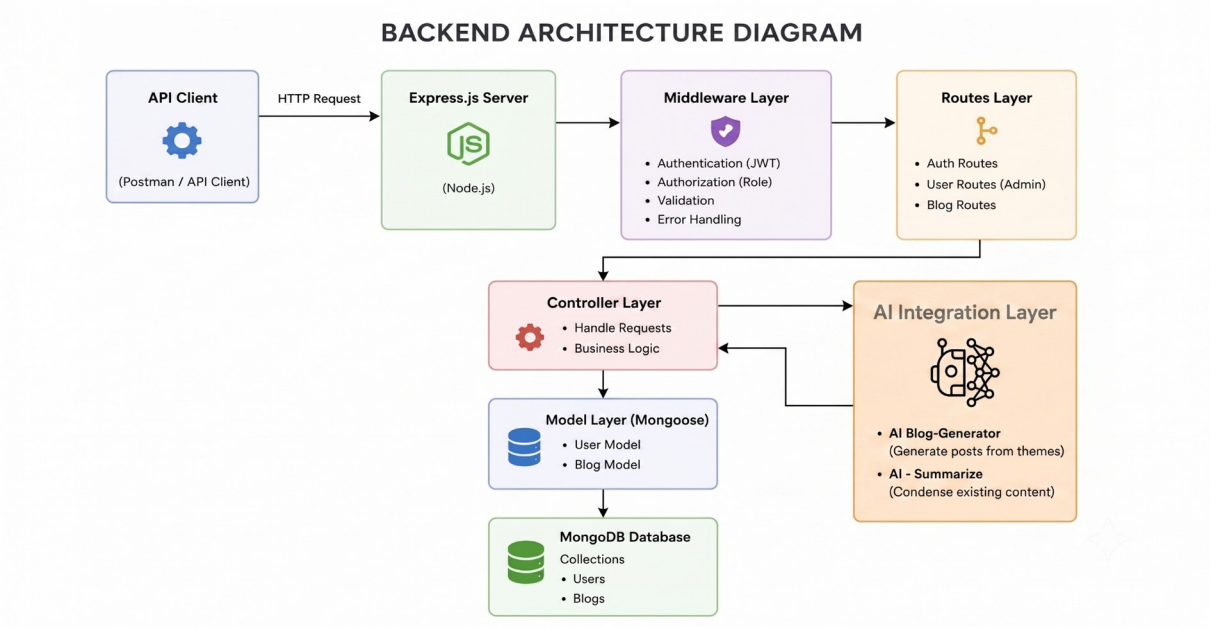



*Figure 1: High-level data flow through AI BlogNest API*

**User Stories**

| User Type | Functional Requirement (Epic) | User Story Number | User Story / Task | Acceptance Criteria | Priority | Release |
|---|---|---|---|---|---|---|
| Author (Registered User) | Registration | USN-1 | As a user, I can register for the | I receive a 201 response | High | Sprint-1 |

| User Type | Functional Requirement (Epic) | User Story Number | User Story / Task | Acceptance Criteria | Priority | Release |
|---|---|---|---|---|---|---|
| | | | application by entering my name, email, and password. | with my new user id, name, and email. | | |
| Author (Registered User) | Login | USN-2 | As a user, I can log into the application using my email & password. | I receive a signed JWT token I can use for authenticated requests. | High | Sprint-1 |
| Author (Registered User) | Profile | USN-3 | As a user, I can view my own profile. | I receive my id, name, and email when authenticated. | Medium | Sprint-1 |
| Author (Registered User) | Blog Management | USN-4 | As an author, I can create a new blog post. | The blog is saved with my user id as the author and returned to me. | High | Sprint-2 |
| Author (Registered User) | Blog Management | USN-5 | As an author, I can update or delete only my own blog posts. | I receive a 403 error if I try to edit a post I did not author. | Medium | Sprint-2 |
| Reader (Public/Guest) | Blog Management | USN-6 | As a reader, I can view all blog posts without an account. | I receive a list of blogs with author name populated. | High | Sprint-2 |
| Reader (Public/Guest) | Blog Management | USN-7 | As a reader, I can view a single blog post by its ID. | I receive the full blog post or a 404 if it doesn't exist. | Medium | Sprint-2 |

| User Type | Functional Requirement (Epic) | User Story Number | User Story / Task | Acceptance Criteria | Priority | Release |
|---|---|---|---|---|---|---|
| Author (Registered User) | AI Content Generation | USN-8 | As an author, I can generate a blog draft from a topic. | A structured blog post is generated, categorized, and saved under my account. | High | Sprint-3 |
| Author (Registered User) | AI Summarization | USN-9 | As an author, I can summarize any block of content. | I receive a short, readable paragraph summarizing the input. | High | Sprint-4 |